Assignment No :10

1. **Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = 2*(L+W) and Area = L*W**

```
KD3_chandrakant_96233>delimiter $$
KD3_chandrakant_96233>create procedure calculate(length float ,breadth float
)
    -> begin
    ->   declare area float;
    -> declare perimeter float;
    -> set area=length*breadth;
    -> set perimeter=2*(length + breadth);
    -> insert into result values(area,perimeter);
    ->
    ->
    -> end;
    -> $$
Query OK, 0 rows affected (0.02 sec)

KD3_chandrakant_96233>delimiter ;
KD3_chandrakant_96233>
KD3_chandrakant_96233>call calculate(3,5);
Query OK, 1 row affected (0.02 sec)

KD3_chandrakant_96233>select * from result;
+------+-----------+
| area | perimeter |
+------+-----------+
|   15 |        16 |
+------+-----------+
1 row in set (0.01 sec)
```

2. **Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.**

```
KD3_chandrakant_96233>delimiter $$
KD3_chandrakant_96233>create procedure computesqcu(p_num INT)
    -> begin
    -> declare v_num INT;
    -> declare square INT;
    -> declare cu INT;
    -> set v_num=p_num;
    ->
    -> set square =v_num * v_num;
    ->  set cu=v_num * v_num * v_num;
    -> insert into temp values(v_num,square,cu);
    -> end;
    -> $$
Query OK, 0 rows affected (0.00 sec)

KD3_chandrakant_96233>delimiter ;
KD3_chandrakant_96233>computesqcu(7);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r 'computesqcu(7)' at line 1
KD3_chandrakant_96233>call computesqcu(7);
Query OK, 1 row affected (0.01 sec)

KD3_chandrakant_96233>select * from temp;
+----------+--------+---------+
| variable | square | varcube |
+----------+--------+---------+
|        7 |     49 |     343 |
+----------+--------+---------+
1 row in set (0.00 sec)

KD3_chandrakant_96233>
```

3. **Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are C= (F-32) *5/9 F= 9/5*C + 32**

```
KD3_chandrakant_96233>delimiter $$
KD3_chandrakant_96233>create procedure tempconversion(p_celtemp FLOAT,p_fartemp FLOAT
    -> begin
    -> declare v_celtemp FLOAT;
    -> declare v_fartemp float;
    ->  declare v_ftc FLOAT;
    -> declare v_ctf FLOAT;
    -> set v_celtemp=p_celtemp;
    -> set v_fartemp=p_fartemp;
    -> set v_ftc=(v_fartemp - 32) * 5/9;
    -> set v_ctf=9/5 * v_celtemp + 32;
    -> insert into temperature values(v_celtemp,v_fartemp,v_ctf,v_ftc);
    -> end;
    -> $$
Query OK, 0 rows affected (0.01 sec)

KD3_chandrakant_96233>delimiter ;
KD3_chandrakant_96233>call tempconversion(25,35);
Query OK, 1 row affected (0.01 sec)

KD3_chandrakant_96233>select * from temperature;
+---------+-------+------+---------+
| celsius | faren | ctf  | ftc     |
+---------+-------+------+---------+
|      25 |    35 |   77 | 1.66667 |
+---------+-------+------+---------+
1 row in set (0.00 sec)
```

4. **Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches**

```
KD3_chandrakant_96233>CREATE PROCEDURE convert_inches(IN total_inches INT)
    -> BEGIN
    ->     DECLARE yards INT;
    ->     DECLARE feet INT;
    ->     DECLARE inches INT;
    ->
    ->     -- Calculate yards
    ->     SET yards = total_inches DIV 36;
    ->
    ->     -- Remaining inches after yards
    ->     SET inches = total_inches MOD 36;
    ->
    ->     -- Calculate feet from remaining inches
    ->     SET feet = inches DIV 12;
    ->
    ->     -- Remaining inches after feet
    ->     SET inches = inches MOD 12;
    ->
    ->     -- Display result
    ->     SELECT
    ->         total_inches AS Total_Inches,
    ->         yards AS Yards,
    ->         feet AS Feet,
    ->         inches AS Inches;
    -> END;
    ->
    ->  $$
Query OK, 0 rows affected (0.01 sec)

KD3_chandrakant_96233>
KD3_chandrakant_96233>DELIMITER ;
KD3_chandrakant_96233>
KD3_chandrakant_96233>call convert_inches(124);
+--------------+-------+------+--------+
| Total_Inches | Yards | Feet | Inches |
+--------------+-------+------+--------+
|          124 |     3 |    1 |      4 |
+--------------+-------+------+--------+
1 row in set (0.00 sec)
```

5. **Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.**

```
KD3_chandrakant_96233>CREATE PROCEDURE check_product(IN num1 DECIMAL(10,2), IN num2 DECIMAL(10,2))
    -> BEGIN
    ->     DECLARE product DECIMAL(10,2);
    ->
    ->     -- Calculate product
    ->     SET product = num1 * num2;
    ->
    ->     -- Check condition
    ->     IF product >= 100 THEN
    ->         SELECT
    ->             num1 AS First_Number,
    ->             num2 AS Second_Number,
    ->             product AS Product,
    ->             'Product is Equal to or Greater than 100' AS Result;
    ->     ELSE
    ->         SELECT
    ->             num1 AS First_Number,
    ->             num2 AS Second_Number,
    ->             product AS Product,
    ->             'Product is Less than 100' AS Result;
    ->     END IF;
    ->
    -> END $$
Query OK, 0 rows affected (0.02 sec)

KD3_chandrakant_96233>
KD3_chandrakant_96233>DELIMITER ;
KD3_chandrakant_96233>call check_product(15,24);
+--------------+---------------+---------+---------------------------------------+
| First_Number | Second_Number | Product | Result                                |
+--------------+---------------+---------+---------------------------------------+
|        15.00 |         24.00 |  360.00 | Product is Equal to or Greater than 100 |
+--------------+---------------+---------+---------------------------------------+
1 row in set (0.01 sec)
```